1 关键公式

$$\boxed{\iiint_{(v)} \left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right) dv = \iint_{(S)} P\,dydz + Q\,dxdy + R\,dxdy}$$

$$\boxed{\begin{aligned} \iiint_{(v)} \frac{\partial P}{\partial x}\,dv &= \iint_{S} P\,dydz \\ \iiint_{(v)} \frac{\partial Q}{\partial y}\,dv &= \iint_{S} Q\,dxdz \\ \iiint_{(v)} \frac{\partial R}{\partial z}\,dv &= \iint_{S} R\,dxdy \end{aligned}}$$

$$\boxed{\int_{(l)} P\,dx + Q\,dy + R\,dz = \iint_{(S)} \left(\frac{\partial Q}{\partial x}-\frac{\partial P}{\partial y}\right) dxdy + \left(\frac{\partial R}{\partial y}-\frac{\partial Q}{\partial z}\right) dydz + \left(\frac{\partial P}{\partial z}-\frac{\partial R}{\partial x}\right) dzdx}$$

2 散度公式

$$\boxed{\iiint_{(v)} \left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right) dv = \iint_{(S)} P\,dydz + Q\,dxdy + R\,dxdy}$$